# Exercises in *Fourier Analysis on Number Fields*

**Solutions to the exercises of Ramakrishnan–Valenza**

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# Contents


1 Topological Groups 3

2 Some Representation Theory 4

3 Duality for Locally Compact Abelian Groups 5

4 The Structure of Arithmetic Fields 6

5 Adeles, Ideles, and the Class Groups 7

6 A Quick Tour of Class Field Theory 8

7 Tate's Thesis and Applications 9

# 1 Topological Groups

# 2 Some Representation Theory

# 3 Duality for Locally Compact Abelian Groups

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# 5 Adeles, Ideles, and the Class Groups

# 6 A Quick Tour of Class Field Theory

# 7 Tate's Thesis and Applications

**Exercise 7.1.** Let $F$ be a local field and let $\psi$ be a nontrivial unitary character on $F$; that is, a continuous, nontrivial homomorphism from $F$ as an additive group into $S^1$, the circle group. For each $a \in F$, define $\psi_a \colon F \to S^1$ by

$$\psi_a(x) = \psi(ax).$$

(a) Show that $\psi_a$ is again a unitary character of $F$, and that $\psi_a$ is trivial if and only if $a = 0$.

(b) Show that $a \mapsto \psi_a$ defines a continuous, injective homomorphism $\alpha_\psi$ of $F$ into its Pontryagin dual $\widehat{F}$.

(c) Show that the image of $\alpha_\psi$ is dense in $\widehat{F}$. [*Hint*: Show that $\psi_a(b) = 1$ for all $a \in F$ if and only if $b = 0$.]

(d) Show that $\alpha_\psi$ is bicontinuous.

(e) Show that $\alpha_\psi(F)$ is a complete, hence closed, subgroup of $\widehat{F}$. Conclude that $\alpha_\psi$ is an isomorphism of topological groups.

*Solution.* (a) Multiplication by $a$, $x \mapsto ax$, is a continuous endomorphism of the additive group $F$. Hence $\psi_a$, its composite with $\psi$, is a continuous homomorphism $F \to S^1$, i.e. a unitary character of $F$.

If $a = 0$, then $\psi_a(x) = \psi(0) = 1$ for all $x \in F$, so $\psi_a$ is trivial. Conversely, let $a \neq 0$. Since $\psi$ is nontrivial, there is some $y \in F$ with $\psi(y) \neq 1$, and then $\psi_a(a^{-1}y) = \psi(y) \neq 1$. So $\psi_a$ is nontrivial.

(b) Recall that $\widehat{F}$ has the compact-open topology. A basis of neighborhoods of a character $\psi_0$ is given by the sets

$$W(\psi_0, C, \varepsilon) = \{\psi \in \widehat{F} : |\psi(x) - \psi_0(x)| < \varepsilon, \forall x \in C\}.$$

where $C \subset F$ is a compact set and $\varepsilon > 0$.

First, reduce to continuity at 0. Note that $\psi_{a+b}(x) = \psi_a(x) \cdot \psi_b(x)$. Then we have that

$$|\psi_a(x) - \psi_b(x)| = |\psi_b| \cdot |\psi_{a-b}(x) - 1| = |\psi_{a-b}(x) - 1|.$$

So it suffices to show that for each compact $C$ and $\varepsilon > 0$, there is a $\delta > 0$, such that $\forall |a| < \delta$, $|\psi_a(x) - 1| < \varepsilon$.

Given an $\varepsilon > 0$. Since $\psi$ is continuous at 0 and $\psi(0) = 1$, then there is an $r > 0$, such that $\forall |x| < r$, $|\psi(x) - 1| < \varepsilon$. Since $C$ is compact, then $M = \max_{x \in C}\{|x|\} < \infty$. Choose $\delta = r/M$, then $\forall |a| < \delta$ and $\forall x \in C$, $|ax| < r$. It follows that

$$|\psi_a(x) - 1| = |\psi(ax) - 1| < \varepsilon.$$

Injectivity is followed from (a).

(c) Note that

$$\forall a \in F, \psi_a(b) = 1 \iff \forall a \in F, \psi(ab) = 1 \iff b = 0.$$

Recall that we have the fact of Pontryagin dual: let $G$ be a locally compact abelian group and $H$ be a proper closed subgroup of $G$. Then there is a nontrivial character $\chi$ of $G$, such that $\chi|_H = 1$.

Let $H$ be the closure of the image of $\alpha_\psi$, i.e. $H = \overline{\alpha_\psi(F)}$. Assume that $H \neq \widehat{F}$. Then we have a character $\chi \in \widehat{\widehat{F}}$ such that $\chi$ is trivial on $H$. And by Pontryagin duality $F \equiv \widehat{\widehat{F}}$, every character of $\widehat{F}$ is evaluation at some point $b \in F$, i.e. $\psi \mapsto \psi(b)$. Then we have that $\chi = (\psi \mapsto \psi(b_0))$ for some $b_0 \in F$. Since $\chi$ is trivial on $H$ then $\forall a \in F, \psi_a(b_0) = 1$. It follows that $b_0 = 0$, which contradicts to $\chi$ is a nontrivial character of $\widehat{F}$.

(d) First, reduce to continuity at $0 \in F$. Note that $\psi_a \in \psi_{a_0} W(1, C, \varepsilon)$ if and only if $\psi_{a-a_0} \in W(1, C, \varepsilon)$. Then it suffices to show that $\forall r > 0$, there are a compact $C$ and $\varepsilon > 0$, such that

$$\psi_a \in W(1, C, \varepsilon) \implies |a| < r.$$

Since $\psi$ is nontrivial, pick $y \in F$ such that $\psi(y) \neq 1$, and set $\varepsilon = |\psi(y) - 1| > 0$. Now choose $C \coloneqq \{x \in F : |x| \leq |y|/r\}$. Since $C$ is closed, then $C$ is compact.

Now suppose that $\psi_a \in W(1, C, \varepsilon)$ but $|a| \geq r$. Then $a \neq 0$ and $x \coloneqq a^{-1}y$ satisfies $|x| = |y|/|a| \leq |y|/r$, so $x \in C$. But

$$|\psi_a(x) - 1| = |\psi(y) - 1| = \varepsilon,$$

which contradicts to $\psi_a \in W(1, C, \varepsilon)$. Therefore, $|a| < r$.

(e) Given a Cauchy sequence $\{\psi_{a_n}\}_n$ in $\alpha_\psi(F)$. Then we have that for every neighborhood $W$ of 1 there is an $N$ such that $\psi_{a_n - a_m} \in W$ for all $n, m \geq N$. By (d), there is a neighborhood $W$ such that

$$\psi_b \in W \implies |b| < r.$$

Then we have that $\{a_n\}_n$ is a Cauchy sequence in $F$. Since $F$ is complete, then $\{a_n\}_n$ converges uniquely to $a \in F$. By (b), we have that $\psi_{a_n}$ converges to $\psi_a$. Since $\widehat{F}$ is Hausdorff, the limit is unique. Therefore, $\alpha_\psi(F)$ is complete. And since every complete subspace of a Hausdorff space is closed, then $\alpha_\psi(F)$ is closed. In conclusion, $\alpha_\psi$ is an isomorphism of topological groups. □

**Exercise 7.2.** Let $F$ be a non-Archimedean local field with ring of integers $\mathfrak{o}_F$ and maximal ideal $P$. Fix a nontrivial unitary character $\psi$ of $F$. Let $P^m$ be the conductor of $\psi$; that is, the largest fractional ideal $P^m$ on which $\psi$ is trivial. Let $\alpha_\psi$, as defined in Exercise 7.1, be the isomorphism between $F$ and its Pontryagin dual.

(a) Show that via $\alpha_\psi$, $P^m$ identifies with the Pontryagin dual of $\widehat{\mathfrak{o}}_F$. (This is also the inverse different of $F$, which soon makes its appearance.)

(b) Let $n \geq 1$ and $x \in F$. Show that $xP^n \subseteq \ker(\psi)$ if and only if $x \in P^{m-n}$.

*Solution.* (a) Claim: $\alpha(P^m)$ is the annilator of $\mathfrak{o}_F$ in $\widehat{F}$, i.e.

$$\alpha(P^m) = \mathfrak{o}_F^\perp \coloneqq \{\chi \in \widehat{F} : \chi|_{\mathfrak{o}_F} = 1\}.$$

In fact, for every $x \in F$,

$$\psi_x \in \mathfrak{o}_F^\perp \iff \psi(x\mathfrak{o}_F) = 1 \iff x\mathfrak{o}_F \subseteq \ker\psi \iff x\mathfrak{o}_F \subseteq P^m \iff x \in P^m.$$

(b) Note that

$$xP^n \subseteq \ker(\psi) \iff xP^n \subseteq P^m \iff v(x) + n \geq m \iff x \in P^{m-n}.$$

□

**Exercise 7.3.** In this exercise we construct nontrivial characters for non-Archimedean local fields, including the standard character for $\mathbb{Q}_p$ and its extensions, and the standard character for the completion of $\mathbb{F}_p(t)$ and its extensions, at the place defined by $t^{-1}$.

(a) Given any $x \in \mathbb{Q}_p$, let $n$ be the smallest nonnegative integer such that $p^n x$ lies in $\mathbb{Z}_p$. Let $r \in \mathbb{Z}$ be such that $r \equiv p^n x \pmod{p^n}$. Put

$$\psi(x) = e^{2\pi i r/p^n}.$$

Show that $\psi\colon \mathbb{Q}_p \to S^1$ is a nontrivial unitary character of conductor $\mathbb{Z}_p$ and that it identifies with the composite map $\mathbb{Q}_p \to \mathbb{Q}_p/\mathbb{Z}_p \to \mathbb{Q}/\mathbb{Z} \to S^1$.

(b) Let $K_\infty$ denote the completion of $K = \mathbb{F}_p(t)$ defined by the place $t^{-1}$. Show that $K_\infty = \mathbb{F}_p(\!(t^{-1})\!)$; that is, that each $x \in K_\infty$ can be uniquely represented by a formal power series

$$\sum_{n=-\infty}^{r} a_n t^n$$

with $a_n \in \mathbb{Z}/p\mathbb{Z}$ for all $n$ less than or equal to the integer $r$. Put

$$\psi_\infty(x) = e^{-2\pi i a_{-1}/p} \in S^1.$$

Show that $\psi_\infty$ is a nontrivial unitary character of $K_\infty$ of conductor $t^{-2}\mathbb{F}_p[\![t^{-1}]\!]$. (*Note*: This differs from the book, which puts $\psi_\infty(x) = e^{2\pi i a_1/p}$, of conductor $\mathbb{F}_p[\![t^{-1}]\!]$; with that choice part (d) of Exercise 7.5 fails.)

(c) Let $\pi$ be any irreducible polynomial in $\mathbb{F}_p[t] \subseteq K$, and let $K_\pi$ denote the completion of $K$ defined by the prime ideal $(\pi)$, with residue field $k$, which we may identify with a transversal of $(\pi)$ in $\mathbb{F}_p[t]$. Show that every $x \in K_\pi$ can be written in the form

$$\sum_{n=r}^{+\infty} a_n \pi^n$$

for some integer $r$, with each coefficient $a_n \in k$. Next put

$$\psi_\pi(x) = e^{2\pi i \operatorname{tr}_{k/\mathbb{F}_p}(a_{-1})/p}.$$

Show that $\psi_\pi$ is a nontrivial unitary character of $K_\pi$.

(d) Let $F$ be any non-Archimedean local field. Show that $F$ is a finite separable extension of a local field $F_0$ as in parts (a), (b), or (c) above. Let $\psi_0$ denote the corresponding character of $F_0$ defined therein. Given $x \in F$, set

$$\psi(x) = \psi_0(\operatorname{tr}_{F/F_0}(x)).$$

Show that $\psi$ is a nontrivial unitary character of $F$.

(e) Continuing in the same context, show that when the characteristic of $F$ is zero, the conductor of $\psi$ is the inverse different $\mathcal{D}_F^{-1}$. In this case show also that for $f$ the characteristic function of $\mathfrak{o}_F$, the Fourier transform of $f$ is $N(\mathcal{D}_F)^{-1/2}$ times the characteristic function of $\mathcal{D}_F^{-1}$.

*Solution.* (a) It is easy to see that $\psi$ is continuous and $|\psi(x)| = 1$. Note that

$$\forall x \in \mathbb{Z}_p,\ \psi(x) = e^{2\pi i r} = 1, i.e.\ \psi|_{\mathbb{Z}_p} = 1$$

And since $\psi(1/p) = e^{2\pi i/p} \neq 1$, then $\mathbb{Z}_p$ is the largest fractional ideal that $\psi$ is trivial.

For any $x_1, x_2 \in \mathbb{Q}_p$, we are going to show $\psi(x_1 + x_2) = \psi(x_1)\psi(x_2)$. Let $n_i$ be the smallest nonnegative integer such that $p^{n_i} x_i$ lies in $\mathbb{Z}_p$, and let $r_i \in \mathbb{Z}$ be such that $r_i \equiv p^{n_i} x_i \pmod{p^{n_i}}$.

WLOG, we can assume that $0 < n_1 < n_2$. Then $n_2$ is the smallest integer such that $p^{n_2}(x+y) \in \mathbb{Z}_p$, and $p^{n_2}(x+y) \equiv p^{n_2-n_1}p^{n_1}x + p^{n_2}y \equiv p^{n_2-n_1}r_1 + r_2 \pmod{p^{n_2}}$. Therefore,

$$\psi(x+y) = e^{2\pi i(p^{n_2-n_1}r_1+r_2)/p^{n_2}} = e^{2\pi i r_1/p^{n_1}} e^{2\pi i r_2/p^{n_2}} = \psi(x)\psi(y).$$

It follows that $\psi$ is a nontrivial unitary character of conductor $\mathbb{Z}_p$. And it identisfied with the composite map $\mathbb{Q}_p \to \mathbb{Q}_p/\mathbb{Z}_p \to \mathbb{Q}/\mathbb{Z} \to S^1$ as follows:

$$x \mapsto p^n x \mapsto r/p^n \mapsto e^{2\pi i r/p^n}.$$

(b) It is easy to see that $\psi_\infty(x+y) = \psi_\infty(x)\psi_\infty(y)$. And for every $x \in t^{-2}\mathbb{F}_p[\![t^{-1}]\!]$, $a_{-1} = 0$, then $\psi_\infty(x) = e^{-2\pi i a_{-1}/p} = 1$. And for $t^{-1} \in t^{-1}\mathbb{F}_p[\![t^{-1}]\!]$, $\psi_\infty(t^{-1}) = e^{-2\pi i/p} \neq 1$. Since every fractional ideal strictly containing $t^{-2}\mathbb{F}_p[\![t^{-1}]\!]$ contains $t^{-1}\mathbb{F}_p[\![t^{-1}]\!]$, the conductor of $\psi_\infty$ is $t^{-2}\mathbb{F}_p[\![t^{-1}]\!]$. In particular, $\psi_\infty$ is trivial on an open subgroup, hence continuous.

(c) Write $A = \mathbb{F}_p[t], d = \deg\pi, q = p^d$. Let $\mathfrak{o}_\pi = \varprojlim A/(\pi^n)$ be the completion of $A$ at $(\pi)$, $K_\pi = \mathfrak{o}_\pi[1/\pi]$ and the residue field $k = \mathfrak{o}_\pi/\pi\mathfrak{o}_\pi = A/(\pi) = \mathbb{F}_q$. Take

$$S = \{f \in A : \deg f < d\}.$$

By division algorithm, we have a bijection between $S$ and $A/(\pi) = k$.

Next we want to show that there is a unique $\pi$-adic expansion. WLOG, we only consider $x \in \mathfrak{o}_\pi$. For general $x \neq 0$, take $r = v(x)$ and apply the result to $y = \pi^{-r}x \in \mathfrak{o}_\pi$.

Since $S \to k$ is surjective, choose $a_0 \in S$ such that $x \equiv a_0 \pmod{\pi}$. Then take $x_1 = (x - a_0)/\pi$ and choose $a_1 \in S$ such that $x_1 \equiv a_1 \pmod{\pi}$. By induction, we have

$$x - \sum_{i=0}^{n-1} a_i\pi^i = \pi^n x_n.$$

Then $|x - \sum_{i=0}^{n-1} a_i\pi^i| \leq q^- n \to 0$. Therefore, $x = \sum_{i\geq 0} a_i\pi^i$. And the uniqueness follows from the irreducibility of $\pi$.

Since $\mathrm{tr}_{k/\mathbb{F}_p}(x+y) = \mathrm{tr}_{k/\mathbb{F}_p}(x) + \mathrm{tr}_{k/\mathbb{F}_p}(y)$, then $\psi_\pi(x+y) = \psi_\pi(x)\psi_\pi(y)$.

(d) Homomorphism property follows from the fact that $\mathrm{tr}_{F/F_0}$ is additive. Since $\mathrm{tr}_{F/F_0}$ is $F_0$ linear on finite dimensional $F_0$-space, then $\mathrm{tr}_{F/F_0}$ is continuous. Since $F/F_0$ is separable extension, then $\mathrm{tr}_{F/F_0}$ is surjective, then $\psi(x)$ is nontrivial.

(e) Since the characteristic of $F$ is 0, then we assume that $F_0 = \mathbb{Q}_p$. Recall that the conductor $\psi_0$ is $\mathbb{Z}_p$ and

$$\mathcal{D}_F^{-1} = \{x \in F : \mathrm{tr}(x\mathfrak{o}) \subseteq \mathbb{Z}_p\}.$$

Claim: For any fractional ideal $\mathfrak{a}$, $\psi|_\mathfrak{a} = 1$ if and only if $\mathfrak{a} \subseteq \mathcal{D}_F^{-1}$.

*Proof of claim.* Note that

$$\psi|_\mathfrak{a} = 1 \iff \psi_0|_{\mathrm{tr}(\mathfrak{a})} = 1 \iff \mathrm{tr}(\mathfrak{a}) \subseteq \mathbb{Z}_p.$$

It remains to show that $\mathrm{tr}(\mathfrak{a}) \subseteq \mathbb{Z}_p \iff \mathfrak{a} \subseteq \mathcal{D}_F^{-1}$.

($\Longleftarrow$): For each $a \in \mathfrak{a} \subseteq \mathcal{D}_F^{-1}$, by definition of $\mathcal{D}_F^{-1}$, we have that $\mathrm{tr}(a) \in \mathbb{Z}_p$.

($\Longrightarrow$): For each $a \in \mathfrak{a}$, $a\mathfrak{o} \subseteq \mathfrak{a}$. It follows that $\mathrm{tr}(a\mathfrak{o}) \subseteq \mathbb{Z}_p$, i.e. $a \in \mathcal{D}_F^{-1}$. □

Next, we will show that

$$\widehat{\mathbb{1}_{\mathfrak{o}_F}} = N(\mathcal{D}_F)^{-1/2}\mathbb{1}_{\mathcal{D}_F^{-1}}.$$

Recall that for $f \in S(F)$, $\widehat{f}(x) = \int_F f(y)\psi(xy)\,\mathrm{d}y$. And we have the follow lemma:

**Lemma 7.1.** *Let $H$ be a compact open subgroup of $F$, $\mu$ a Haar measure on $F$, and $\chi$ a character of $H$. Then*

$$\int_H \chi\,\mathrm{d}\mu = \begin{cases} \mu(H), & \text{if } \chi = 1, \\ 0, & \text{otherwise.} \end{cases}$$

*Sketch of proof.* If $\chi$ is nontrivial, then there is a $y \in H$ such that $\chi(y) \neq 1$, then $\int_H \chi(x)\,\mathrm{d}\mu(x) = \int_H \chi(xy)\,\mathrm{d}\mu(xy) = \chi(y)\int_H \chi(x)\,\mathrm{d}\mu(x)$. Hence, $\int_H \chi(x)\,\mathrm{d}\mu(x) = 0$. □

Let $\mathrm{d}x$ be the Haar measure such that $\int_{\mathfrak{o}} \mathrm{d}x = 1$. Note that $\psi(xy)$ is trivial on $\mathfrak{o}$ if and only if $x \in \mathcal{D}_F^{-1}$.

$$\widehat{\mathbb{1}_{\mathfrak{o}_F}}(x) = \int_{\mathfrak{o}} \psi(xy)\,\mathrm{d}y = \int_{\mathfrak{o}} \psi_0(\mathrm{tr}(xy))\,\mathrm{d}y = \mathbb{1}_{\mathcal{D}_F^{-1}}. \tag{7.1}$$

Assume that $c'\,\mathrm{d}x$ be the dual measure of $\mathrm{d}x$ under the transport $\alpha$. Fourier inversion formula should holds for $f = \mathbb{1}_{\mathfrak{o}}$, i.e. $\widehat{\widehat{\mathbb{1}_{\mathfrak{o}}}} = \mathbb{1}_{\mathfrak{o}}$. Note that

$$\widehat{\widehat{\mathbb{1}_{\mathfrak{o}}}}(x) = \int_F \mathbb{1}_{\mathcal{D}_F^{-1}}(y)\psi(-xy)c'\,\mathrm{d}y = c'\int_{\mathcal{D}_F^{-1}} \psi(-xy)\,\mathrm{d}y = c'\,\mathrm{vol}(\mathcal{D}_F^{-1}, \mathrm{d}x)\mathbb{1}_{\mathfrak{o}}.$$

And note that $\mathrm{vol}(\mathcal{D}_F^{-1}, \mathrm{d}x) = N(\mathcal{D}_F)$. It follows that $c' = N(\mathcal{D}_F)^{-1}$.

Now assume that for the self-dual has the form $t\,\mathrm{d}x$, then the dual Haar measure, transported by $\alpha$ is $t^{-1}c'\,\mathrm{d}x = t^{-1}N(\mathcal{D}_F)^{-1}\,\mathrm{d}x$. Then we have the following equality:

$$t\,\mathrm{d}x = t^{-1}N(\mathcal{D}_F)^{-1}\,\mathrm{d}x.$$

It follows that $t = N(\mathcal{D}_F)^{-1/2}$. Normalize the equation (7.1), we complete this exercise. □

**Exercise 7.4.** We shall now construct the standard character on $\mathbb{A}_K/K$ for an algebraic number field $K$.

(a) For each rational prime $p$, let $\psi_p$ denote the standard character of $\mathbb{Q}_p$ defined above. Also define $\psi_\infty \colon \mathbb{R} \to S^1$ by $\psi_\infty(x) = e^{-2\pi i x}$. Show that

$$\psi \colon \mathbb{A}_{\mathbb{Q}} \to S^1, \qquad (x_v) \mapsto \prod_v \psi_v(x_v)$$

is a nontrivial unitary character on $\mathbb{A}_{\mathbb{Q}}$, which is trivial on $\mathbb{Q}$ (embedded diagonally). [*Hint*: Use the Artin product formula.]

(b) For any number field $K$, define $\psi_K \colon \mathbb{A}_K \to S^1$ by

$$\psi_K(x) = \prod_v \psi_v(x_v)$$

where $v$ ranges over all the places of $K$ and $\psi_v$ is the standard character on the local field $K_v$, as constructed above. Show that $\psi_K$ is a nontrivial unitary character on $\mathbb{A}_K$.

(c) Show also that $\psi_K(x) = \psi(\mathrm{tr}(x))$ where tr denotes the adelic trace map from $K$ to $\mathbb{Q}$. Conclude from part (a) that $\psi_K$ is trivial on $K$.

*Solution.* (a) Recall that

**Theorem 7.2** (Artin product formula)**.** *For a global field $K$, namely number field or function field of a curve over $\mathbb{F}_q$, and any non-zero $x \in K$, we have that*

$$\prod_v |x|_v = 1,$$

*where $v$ run over all places of $K$.*

It is easy to check that $\psi(x+y) = \psi(x)\psi(y)$. Since for any place $v$, $|\psi_v(x_v)| = 1$, then $|\psi(x)| = 1$. Note that: a homomorphism of topological groups that is continuous on a open subgroup is continuous everywhere. It suffices to show that $\psi$ is continuous on some open subgroup $U$. Take $U = \mathbb{R} \times \prod_p \mathbb{Z}_p$. Then $\psi(x) = e^{-2\pi i x_\infty}$, which is continuous.

Note that as a additive group $\mathbb{Q}$ is generated by 1 and the numbers $1/p^k, k \geq 1$. Then it suffices to check that for all prime number $p$ and positive $k$, $\psi(1/p^k) = 1$. In fact,

$$\psi(1/p^k) = \psi_p(1/p^k)\psi_\infty(1/p^k) = e^{2\pi i/p^k} e^{-2\pi i/p^k} = 1.$$

(b) It follows from similar discussion above.

(c) Recall that $\mathrm{tr} : \mathbb{A}_K \to \mathbb{A}_\mathbb{Q}$ is defined componentwise by

$$\mathrm{tr}(x)_p = \sum_{v|p} \mathrm{tr}_{K_v/\mathbb{Q}_p}(x_v), \quad \mathrm{tr}(x)_\infty = \sum_{v|\infty} \mathrm{tr}_{K_v/\mathbb{R}}(x_v).$$

For each $\psi_p$ is additive and there is only finitely many places $v$ lie over $p$. Then we have

$$\psi(\mathrm{tr}_{K/\mathbb{Q}}(x)) = \prod_{p\leq\infty} \psi_p(\sum_{v|p}(\mathrm{tr}_{K_v/\mathbb{Q}_p}(x_v))) = \prod_{p\leq\infty}\prod_{v|p} \psi_p(\mathrm{tr}_{K_v/\mathbb{Q}_p}(x_v)) = \prod_v \psi_v(x_v) = \psi_K(x).$$

By (a), we have that for all $y \in \mathbb{Q}$, $\psi(y) = 1$ and for all $x \in K$, $\mathrm{tr}_{K/\mathbb{Q}}(x) \in \mathbb{Q}$, then $\psi_K(x) = \psi(\mathrm{tr}_{K/\mathbb{Q}}(x)) = 1$. □

**Exercise 7.5.** We shall now construct the standard character for the completion $K_v$ of $K = \mathbb{F}_p(t)$ at any *finite* place $v$.

(a) Let $\pi(t)$ be an irreducible element of $\mathbb{F}_p[t]$ of degree $d \geq 1$. Show that $K$ has a unique place $v$ such that the polynomial $\pi(t)$ generates the maximal ideal of $\mathfrak{o}_v = \{\alpha \in K_v : |\alpha|_v \leq 1\}$. Show further that $q_v = \mathrm{Card}(\mathfrak{o}_v/\pi\mathfrak{o}_v) = p^d$, and that the polynomials in $\mathbb{F}_p[t]$ of degree less than $d$ in fact generate $\mathfrak{o}_v/\pi\mathfrak{o}_v$.

(b) Given any place $v$ of $K$ (possibly infinite), let us define

$$K^{(v)} = \{x \in K : |x|_u \leq 1 \ \forall u \neq v\}.$$

Show that $K_v = K^{(v)} \oplus \mathfrak{o}_v$. [*Hint*: By replacing $t$ by $t^{-1}$ if necessary, one may assume that $v$ is defined by an irreducible polynomial $\pi(t) \in \mathbb{F}_p[t]$ and then apply the previous part.]

(c) Let $v$ be any finite place of $K$; that is, one defined by an irreducible polynomial $\pi(t) \in \mathbb{F}_p[t]$ of positive degree $d$. Show that every element $y$ of $K^{(v)}$ can be written as $z/\pi^n$ for some $n \geq 0$ and polynomial $z = z(t) \in \mathbb{F}_p[t]$ of the form

$$c_1 t^{nd-1} + c_2 t^{nd-2} + \cdots + c_{nd}$$

with $c_j \in \mathbb{F}_p$ for all $j$, possibly all zero. If $x = y + \alpha \in K_v$ with $\alpha \in \mathfrak{o}_v$, put

$$\psi_v(x) = e^{2\pi i c_1/p}.$$

(This is of course well-defined despite the status of $c_1$ as a residue class modulo the prime $p$.) Show that $\psi_v$ is a nontrivial character of $K_v$ of conductor $\mathfrak{o}_v$.

(d) Let $v$, $\pi$, and $y$ be as in the previous part. Show that $y \equiv c_1 t^{-1} \pmod{t^{-2}}$. Conclude that $\psi_v(y)\psi_\infty(y) = 1$, where $\psi_\infty$ is as in part (b) of Exercise 7.3. [*Hint*: Write $\pi$ as $t^d g$, with $g$ a polynomial in $t^{-1}$ of constant term 1, so that $y = zg^{-n}t^{-nd}$.]

*Solution.* Let $A = \mathbb{F}_p[t]$. We consider $\pi$ to be a monic polynomial.

(a) Existence: take $v = v_\pi$.

Uniqueness: suppose that $|\pi|_v < 1$.

- Case (1): $|t|_v > 1$. For $f = \sum_{i=0}^m a_i t^i$ with $a_m \neq 0$, the terms have distinct absolute values $|t|_v^i$. Therefore, $|f|_v = |t|_v^m$. and in particular $|\pi|_v = |t|_v^d > 1$. This is a contradiction.
- Case (2): $|t|_v \leq 1$. By triangle inequality, $A \subseteq \mathfrak{o}_v$, and $P = \{f \in A : |f|_v < 1\}$ is a proper prime ideal containing $\pi$. Since $(\pi)$ is maximal, then $P = (\pi)$. Write $f = \pi^k g$ with $\pi \nmid g$. Then $|g|_v = 1$, so $|f|_v = |\pi|_v^{\operatorname{ord}_\pi f}$. Hence, place $v$ is equivalent to $v_\pi$.

As we know that $\mathfrak{o}_v/\pi\mathfrak{o}_v \cong A/(\pi)$. Then we have that $\operatorname{Card} \mathfrak{o}_v/\pi\mathfrak{o}_v = p^d$.

(b) Note that $K^{(v)} \cap \mathfrak{o}_v = \{x \in K : |x|_u \leq 1, \forall u\}$. Such an $x$ does not have finite poles, then $x \in A$. And since $|x|_\infty \leq 1$, then $\deg x \leq 0$. Then $K^{(v)} \cap \mathfrak{o}_v = \mathbb{F}_p$, but not 0. Therefore the sum $K_v = K^{(v)} + \mathfrak{o}_v$ is **not direct sum**. It can be written as a direct sum as follow,

$$K_v = K_0^{(v)} \oplus \mathfrak{o}_v,$$

where $K_0^{(v)} = \{y \in K^{(v)} : |y|_\infty < 1\}$ for $v$ finite, and $K_0^{(\infty)} = t\mathbb{F}_p[t]$. In both cases $K_0^{(v)} \cap \mathbb{F}_p = 0$, hence $K_0^{(v)} \cap \mathfrak{o}_v = 0$ and the sum is direct.

Finite $v$. Let $x \in K_v$. Then $x$ can be written as

$$x = \sum_{n \geq r} a_n \pi^n = \sum_{n=r}^{-1} a_n \pi^n + \sum_{n=0}^{\infty} a_n \pi^n.$$

Note that $\sum_{n=r}^{-1} a_n \pi^n \in K_0^{(v)}$ and $\sum_{n=0}^{\infty} a_n \pi^n \in \mathfrak{o}_v$.

Infinity $v$. Recall that $K_\infty = \mathbb{F}_p(\!(t^{-1})\!)$ and $x \in K_\infty$ can express as

$$x = \sum_{n=-\infty}^{r} a_n t^n = \sum_{n=1}^{r} a_n t^n + \sum_{n=-\infty}^{0} a_n t^n.$$

And $\mathfrak{o}_\infty = \mathbb{F}_p[\![t^{-1}]\!]$ and $K^{(\infty)} = \mathbb{F}_p[t]$. Note that $\sum_{n=1}^{r} a_n t^n \in t\mathbb{F}_p[t] = K_0^{(\infty)}$ and $\sum_{n \leq 0} a_n t^n \in \mathfrak{o}_\infty$. Then we have that $K_\infty = K_0^{(\infty)} \oplus \mathfrak{o}_\infty$. (Again $K^{(\infty)} \cap \mathfrak{o}_\infty = \mathbb{F}_p$, so $K^{(\infty)} + \mathfrak{o}_\infty$ is not direct.)

(c) In this part of exercise, we will describe the form of the elements of $K^{(v)}$.

Let $y \in K^{(v)}$ and write $y = a/b$ with $a, b$ coprime and $b$ monic. If there is another prime $\varpi \neq \pi$ divided $b$, then $|y|_\varpi > 1$, which contradicts to $y \in K^{(v)}$. So $b = \pi^n$. And since $|y|_\infty \leq 1$, then $\deg a \leq nd$. If $\deg a = nd$, then we can split off the constant as follows:

$$y = c + (a - c\pi^n)/\pi^n,$$

where $c \in \mathbb{F}_p \subseteq \mathfrak{o}_v$ is the leading coefficient of $a$.

According to above discuss, we have that $y \in K^{(v)}$ can be express as $z/\pi^n$ with $\deg z < nd$. The coefficient $c_1$ does not depend on the choice of $n$: $z/\pi^n = z\pi/\pi^{n+1}$, and since $\pi$ is monic, the coefficient of $t^{(n+1)d-1}$ in $z\pi$ equals the coefficient of $t^{nd-1}$ in $z$. Together with the uniqueness of the decomposition $x = y + \alpha$ in (b), this shows that $\psi_v$ is well defined. Writing two elements of $K_0^{(v)}$ with the same $n$, we see that $c_1$ is additive, so $\psi_v$ is additive. Clearly $|\psi_v(x)| = 1$ for all $x$. Note that for $x \in \mathfrak{o}_v$ we have that $y = 0$, then $\psi_v|_{\mathfrak{o}_v} = 1$ i.e. $\psi_v$ is trivial on the open subgroup $\mathfrak{o}_v$. Then $\psi_v$ is continuous. Hence, $\psi_v$ is a unitary character.

Note that for $x = t^{d-1}/\pi \in \pi^{-1}\mathfrak{o}_v$, $\psi_v(x) = e^{2\pi i/p} \neq 1$. Hence, the conductor of $\psi_v$ is $\mathfrak{o}_v$.

(d) Since $\pi$ is monic, $\pi = t^d g$, where $g = 1 + b_1 t^{-1} + \cdots + b_d t^{-d}$. Hence, $g \in \mathbb{F}_p[\![t^{-1}]\!]^\times$ and $g^{-n} \equiv 1 \pmod{t^{-1}}$. Therefore,

$$y = \frac{z}{\pi^n} = zt^{-nd} g^{-n} = (c_1 t^{-1} + c_2 t^{-2} + \cdots + c_{nd} t^{-nd})(1 + O(t^{-1})) \equiv c_1 t^{-1} \pmod{t^{-2}}.$$

By the definition of $\psi_\infty(y) = e^{-2\pi i c_1/p}$ in part (b) of Exercise 7.3, then we have that $\psi_v(y)\psi_\infty(y) = 1$. □

**Exercise 7.6.** We may now use the results of previous problems to construct the standard character $\psi_K \colon \mathbb{A}_K \to S^1$ for $K$ any global field of positive characteristic.

(a) Suppose that $K_0 = \mathbb{F}_p(t)$. Define $\psi_{K_0}$ by

$$\psi_{K_0}(x) = \prod_v \psi_v(x_v)$$

where $\psi_\infty$ is given by Exercise 7.3 and $\psi_v$ ($v$ finite) is given by part (c) of the previous exercise. Show that $\psi_{K_0}$ is a nontrivial character of $\mathbb{A}_{K_0}$ that, moreover, is trivial on $K_0$.

(b) Now let $K$ be any finite separable extension of $K_0$. Use the trace map from $K$ to $K_0$ (or, equivalently, the corresponding adelic trace map) to define the standard character $\psi_K$ on $K$. Show that this character is nontrivial on $\mathbb{A}_K$, but trivial on $K$.

*Solution.* (a) Let $U = K_\infty \times \prod_v \mathfrak{o}_v$. Note that $\psi_{K_0}(x)|_U = \psi_\infty(x)$, and $U$ is an open subgroup of $\mathbb{A}_{K_0}$. And for element $x = (1/t, 0, 0, \ldots, 0) \in \mathbb{A}_{K_0}$, $\psi_{K_0}(x) = \psi_\infty(1/t) = e^{2\pi i/p}$. Then $\psi_{K_0}$ is nontrivial.

Claim: for every $x \in K_0$, $x$ can be written as

$$x = f + \sum_{i=0}^{r} \frac{z_i}{\pi_i^{n_i}},$$

where $f \in \mathbb{F}_p[t]$ and $\deg z_i < n_i d_i$, $d_i = \deg \pi_i$.

*Proof of claim:* let $x = a/b$, where $b = \prod_{i=1}^r \pi_i^{n_i}$, where $\pi_i$ are different monic irreducible polynomial. Let $b_i = b/\pi_i^{n_i}$ and these are coprime. Since $\mathbb{F}_p[t]$ is principal ideal domain, then there exist $u_i{}_{i=1}^r$ such that $\sum_{i=1}^r u_i b_i = 1$. Hence $x = \sum_{i=1}^r a u_i/\pi_i^{n_i}$. By division algorithm, we have that $au_i = q_i \pi_i^{n_i} + z_i$, where $\deg z_i < n_i \deg \pi_i$.

Since $\psi_{K_0}$ is a homomorphism, then we only need to check that for $f \in \mathbb{F}_p[t]$ and $y = z/\pi^n$ satisfy $\psi_{K_0}(f) = 1$ and $\psi_{K_0}(y) = 1$.

- For $f \in \mathbb{F}_p[t]$, $f \in \mathfrak{o}_v$ for all finite place $v$. It follows that $\psi_v(f) = 1$. And at the infinite place $\infty$, $\psi_\infty(f) = 1$, since $a_{-1} = 0$.
- For $y = z/\pi^n$ with $\deg z < nd$ and let $y \equiv c_1 t^{-1} \pmod{t^{-2}}$. For $v = v_\pi$, $\psi_v(y) = e^{2\pi i c_1/p}$. And for any other finite place $u \neq v$, $\psi_u(y) = 1$. For infinite place $\infty$, $\psi_\infty(y) = e^{-2\pi i c_1/p}$.

In conclusion, $\forall x \in K_0$, $\psi_{K_0}(x) = 1$, i.e. $\psi_{K_0}|_{K_0} = 1$.

(b) Recall that adelic tract is defined as follows:

$$\mathrm{tr}(x)_v = \sum_{w|v} \mathrm{tr}_{K_w/K_{0,v}}(x_w).$$

We define $\psi_K = \psi_{K_0} \circ \mathrm{tr}$ or equivalently $\psi_K(x) = \prod_w \psi_w(x_w)$, where $\psi_w = \psi_v \circ \mathrm{tr}_{K_w/K_{0,v}}$. Due to separability of $K/K_0$, then $\mathrm{tr}_{K/K_0}$ is nontrivial and hence $\psi_K$ is non-trivial.

Claim: for any $a \in K$ and every place $v$, then $\mathrm{tr}_{K/K_0}(a) = \sum_{w|v} \mathrm{tr}_{K_w/K_{0,v}}(a)$ This means that adelic trace restricts to $K$ is global trace, that is $\mathrm{tr}\,|_K = \mathrm{tr}_{K/K_0}$.

*Proof of claim:* since $K/K_0$ is separable, $K \otimes_{K_0} K_{0,v} \cong \prod_{w|v} K_w$. The matrix of multiplication by $a$ map in $K$ related to $K_0$-basis $\{e_i\}$ is the same as in $K \otimes K_{0,v}$ as $K_{0,v}$-basis $\{e_i \otimes 1\}$. Then

$$\mathrm{tr}_{K/K_0}(a) = \mathrm{tr}_{(K \otimes K_{0,v})/K_{0,v}}(a \otimes 1) = \sum_w \mathrm{tr}_{K_w/K_{0,v}}(a).$$

Then $\psi_K(a) = \psi_{K_0}(\mathrm{tr}_{K/K_0}\, a)$. It follows that $\mathrm{tr}_{K/K_0}(a) \in K_0$. By $(a)$, $\psi_K(a) = 1$.

□

**Exercise 7.7.** Let $G$ be a locally compact abelian group with Haar measure $\mathrm{d}g$, and let $\mathrm{d}\widehat{g}$ be the dual measure; that is, the measure on the Pontryagin dual $\widehat{G}$ relative to which the Fourier inversion formula holds. Suppose that we have an isomorphism $\alpha\colon G \xrightarrow{\cong} \widehat{G}$ of topological groups.

(a) Show that there is a *unique* multiple $\mu = t \cdot \mathrm{d}g$ such that $\mu$ identifies with its dual measure under $\alpha$. One calls $\mu$ the *self-dual* measure on $G$ relative to the isomorphism $\alpha$.

(b) Let $G$ be the additive group of a non-Archimedean local field $F$, $\psi\colon F \to S^1$ the standard unitary character, and $\alpha_\psi$ the isomorphism of Exercise 7.1. Denote by $\mathrm{d}x$ the self-dual measure relative to $\psi$ (that is, relative to $\alpha_\psi$). Show that $\mathrm{vol}(\mathfrak{o}_F, \mathrm{d}x) = N(\mathcal{D}_F)^{-1/2}$, where $\mathcal{D}_F$ denotes the different of $F$.

*Solution.* □

**Exercise 7.8.** In this exercise we analyze the dependence of the epsilon factor on the additive character $\psi$ and the Haar measure $\mathrm{d}x$. Let $F$ be a local field and consider $\varepsilon(\chi, \psi, \mathrm{d}x)$ for any $\chi \in \mathrm{Hom}_{\mathrm{cont}}(F^\times, \mathbb{C}^\times)$.

(a) For every positive real number $t$, show that

$$\varepsilon(\chi, \psi, t \cdot \mathrm{d}x) = t \cdot \varepsilon(\chi, \psi, \mathrm{d}x).$$

(b) Let $a \in F^\times$, and let $\psi_a$ denote the character defined by $\psi_a(x) = \psi(ax)$. Show that

$$\varepsilon(\chi, \psi_a, \mathrm{d}x) = \chi(a)|a|^{-1}\varepsilon(\chi, \psi, \mathrm{d}x).$$

(c) Let $F$ be non-Archimedean with unique prime ideal $P$, and let $P^n$ and $P^m$ be the conductors of $\chi$ and $\psi$, respectively. Then show for every unramified character $\nu$ of $F^\times$ that

$$\varepsilon(\chi\nu, \psi, \mathrm{d}x) = \nu(\pi^{n-m})\varepsilon(\chi, \psi, \mathrm{d}x)$$

where $\pi$ is a uniformizing parameter for $\mathfrak{o}_F$. Note that the first factor on the right is well-defined because $\nu$ is unramified. [*Hint*: Use the explicit formula given in [RV99, Eq. (7.5)].] (*Note*: This differs from the book, which has $\nu(\pi^{m+n})$, inconsistent with part (b). Also, the book's hint refers to [RV99, Eq. (7.6)], but the relevant formula is [RV99, Eq. (7.5)].)

*Solution.* Recall that

$$\varepsilon(\chi, \psi, \mathrm{d}x) = \frac{L(\chi)Z(\widehat{f}, \chi^\vee)}{L(\chi^\vee)Z(f, \chi)},$$

where $\chi^\vee = \chi^{-1}|\cdot|$, $Z(f, \chi) = \int_{F^\times} f(x)\chi(x)\,\mathrm{d}x$.

(a) Let $\widehat{f}_t$ be the fourier transform of $f$ with respect to the measure $t\,\mathrm{d}x$. Then

$$\widehat{f}_t(x) = \int_F f(y)\psi(xy)t\,\mathrm{d}x = t\widehat{f}(x).$$

It follows that $Z(\widehat{f}_t, \chi^\vee) = tZ(\widehat{f}, \chi^\vee)$. Since $L(\chi)$, $L(\chi^\vee)$ is independent on measure, then

$$\varepsilon(\chi, \psi, t\,\mathrm{d}x) = \frac{L(\chi)Z(\widehat{f}_t, \chi^\vee)}{L(\chi^\vee)Z(f, \chi)} = \frac{L(\chi)t \cdot Z(\widehat{f}, \chi^\vee)}{L(\chi^\vee)Z(f, \chi)} = t \cdot \varepsilon(\chi, \psi, \mathrm{d}x).$$

(b) Let $\widehat{f_a}$ be the fourier transform of $f$ with respect to the character $\psi_a$. Note that

$$\widehat{f_a}(x) = \int_F f(y)\psi_a(xy)\,\mathrm{d}y = \int_F f(y)\psi(axy)\,\mathrm{d}y = \widehat{f}(ax).$$

Then

$$Z(\widehat{f_a},\chi^\vee)=\int_{F^\times}\widehat{f_a}(x)\chi^\vee(x)\,\mathrm{d}^\times x=\chi^\vee(a)^{-1}\int_{F^\times}\widehat{f}(ax)\chi^\vee(ax)\,\mathrm{d}^\times ax=\chi^\vee(a)^{-1}Z(\widehat{f},\chi^\vee).$$

It follows that

$$\varepsilon(\chi,\psi_a,\mathrm{d}x)=\chi^\vee(a)^{-1}\varepsilon(\chi,\psi,\mathrm{d}x)=\chi(a)|a|^{-1}\varepsilon(\chi,\psi,\mathrm{d}x).$$

(c) □

**Exercise 7.9.** Let $F$ be a local field with standard character $\psi$ and self-dual measure $\mathrm{d}x$, and let $\chi=\omega|\cdot|^s$ be a (quasi-)character on $F^\times$ with $\omega$ unitary. Put

$$W(\omega)=\varepsilon(\chi,\psi,\mathrm{d}x)\big|_{s=1/2}.$$

(a) By using the functional equation, show that

$$W(\omega)W(\overline{\omega})=\omega(-1).$$

Conclude that $|W(\omega)|=1$. [*Hint*: First prove that $W(\overline{\omega})=\omega(-1)\overline{W(\omega)}$.]

(b) Using [RV99, Section 7.1], show that for $F=\mathbb{R}$, $W(\omega)=1$ (respectively, $-i$) for $\omega=1$ (respectively, $\omega=\mathrm{sgn}$). Show further that when $F=\mathbb{C}$, we have $W(\omega)=1$.

(c) Let $F$ be non-Archimedean with uniformizing parameter $\pi$, and let $\omega$ as above have conductor $\pi^n\mathfrak{o}_F$. Put

$$G=\sum_{x\in U/U_n}\overline{\omega}(x)\psi(x\pi^{-d-n})$$

where $\pi^d\mathfrak{o}_F=\mathcal{D}_F$ (the different), $U_r=1+\pi^r\mathfrak{o}_F$ ($r>0$), and $U$ is the full group of units in $\mathfrak{o}_F$. Show directly that $|G|^2=q^n$. Then appeal to [RV99, Eq. (7.6)] to deduce (once again) that $|W(\omega)|=1$.

*Solution.* (a) Recall that we have the functional equation:

$$\varepsilon(\chi,\psi,\mathrm{d}x)=\frac{L(\chi)Z(\widehat{f},\chi^\vee)}{L(\chi^\vee)Z(f,\chi)}.$$

And note that

$$(\omega|\cdot|^{1/2})^\vee=\overline{\omega}|\cdot|^{1/2}.$$

Then we have that

$$W(\omega)=\frac{L(\omega|\cdot|^{1/2})Z(\widehat{f},\overline{\omega}|\cdot|^{1/2})}{L(\overline{\omega}|\cdot|^{1/2})Z(f,\chi)}. \tag{7.2}$$

And apply $f=\widehat{f}$ and $\chi=\overline{\omega}|\cdot|^{1/2}$ at the functional equation, i.e.

$$W(\overline{\omega})=\frac{L(\overline{\omega}|\cdot|^{1/2})Z(\widehat{\widehat{f}},\omega|\cdot|^{1/2})}{L(\omega|\cdot|^{1/2})Z(\widehat{f},\overline{\omega}|\cdot|^{1/2})}. \tag{7.3}$$

Since $\widehat{\widehat{f}}(x)=f(-x)$, then for any $\chi$,

$$Z(\widehat{\widehat{f}},\chi)=\int_{F^\times}f(-x)\chi(x)\,\mathrm{d}^\times x=\int_{F^\times}f(-x)\chi(-x)\chi(-1)\,\mathrm{d}^\times(-x)=\chi(-1)Z(f,\chi).$$

Multiplying equation (7.2) and equation (7.3), we have that

$$W(\omega)W(\overline{\omega})=\frac{Z(\widehat{\widehat{f}},\omega|\cdot|^{1/2})}{Z(f,\omega|\cdot|^{1/2})}=\omega(-1). \tag{7.4}$$

**Lemma 7.3.** *For any $\chi$ and $\psi$, we have that*

$$\overline{\varepsilon(\chi,\psi,\mathrm{d}x)}=\varepsilon(\overline{\chi},\overline{\psi},\mathrm{d}x).$$

*Proof of lemma.* TODO. □

Note that $\psi$ is a unitary character, then $\overline{\psi}=\psi_{-1}$. By exercise 7.8 (b), note that for any $\chi$,

$$\varepsilon(\chi,\overline{\psi},\mathrm{d}x)=\varepsilon(\chi,\psi_{-1},\mathrm{d}x)=\chi(-1)\varepsilon(\chi,\psi,\mathrm{d}x).$$

Then

$$\overline{W(\omega)}=\overline{\varepsilon(\omega|\cdot|^{1/2},\psi,\mathrm{d}x)}=\varepsilon(\overline{\omega}|\cdot|^{1/2},\overline{\psi},\mathrm{d}x)=\omega(-1)\varepsilon(\overline{\omega}|\cdot|^{1/2},\psi,\mathrm{d}x)=\omega(-1)W(\overline{\omega}).$$

And by equation (7.4), we have that

$$|W(\omega)|^2=W(\omega)\overline{W(\omega)}=W(\omega)W(-1)W(\overline{\omega})=W(-1)W(-1)=1.$$

(b) It follows from the calculation in section 7.1.
(c) Denote that $\varphi(x)=\psi(\pi^{-d-n}x)$, then $G=\sum_{x\in U/U_n}\overline{\omega}(x)\varphi(x)$. Note that

$$|G|^2=G\overline{G}=\sum_{x,y\in U/U_n}\overline{\omega}(xy^{-1})\varphi(x-y)=\sum_{t\in U/U_n}\overline{\omega}(t)H(t),$$

where $t=xy^{-1}$ and $H(t)=\sum_{y\in U/U_n}\varphi(y(t-1))$.
Note that $U/U_n\cong(\mathfrak{o}_F/P^n)^\times$. Now let $a=t-1$, then $H(t)=\sum_{y\in U/U_n}\varphi(ya)$. Since the conductor of $\psi$ is $\mathcal{D}_F^{-1}$, then $\varphi$ is trivial on $P^n$, i.e. $\varphi$ is a character over $\mathfrak{o}_F/P^n$. And $\varphi(ya)$ only depends on $y,a \mod P^n$. Then we could rephrase the sum $H(t)$ as follow:

$$H(t)=\sum_{y\in\mathfrak{o}_F/P^n}\varphi(ya)-\sum_{y\in P/P^n}\varphi(ya).$$

By the fact that if the character $\varphi_a=(y\mapsto\varphi(ya))$ is trivial, then

$$\sum_{y\in\mathfrak{o}_F/P^n}\varphi(ya)=\#(\mathfrak{o}_F/P^n)=q^n,$$

otherwise the sum is zero.
Note that

$$\varphi_a|_{\mathfrak{o}_F}=1\iff v(a)\geq n\iff a\in P^n\iff t\in U_n,$$

and

$$\varphi_a|_P=1\iff v(a)+1\geq n\iff a\in P^{n-1}\iff t\in U_{n-1}.$$

Then $H(t)=q^n\mathbb{1}_{U_n}-q^{n-1}\mathbb{1}_{U_{n-1}}$. It follows that

$$|G|^2=\sum_{t\in U/U_n}\overline{\omega}(t)H(t)=q^n\overline{\omega}(1)-q^{n-1}\sum_{t\in U_{n-1}/U_n}\overline{\omega}(t).$$

Since $\omega$ is of conductor $\pi^n\mathfrak{o}_F$, then $\omega$ is nontrivial on $U_{n-1}$. It follows that

$$\sum_{t\in U_{n-1}/U_n}\overline{\omega}(t)=0.$$

In conclusion, $|G|^2=q^n\overline{\omega}(1)=q^n$.
Recall that equation (7.6) is

$$W(\omega)=q^{-n/2}\sum_{x\in U/U_n}\overline{\omega}(x)\psi(x\pi^{-d-n})=q^{-n/2}G.$$

We conclude that

$$|W(\omega)|^2 = W(\omega)\overline{W(\omega)} = q^{-n/2}G \cdot q^{-n/2}\overline{G} = q^{-n} \cdot q^n = 1.$$

□

**Exercise 7.10.** (Tate) Let $F$ be a non-Archimedean local field with uniformizing parameter $\pi$, and let $\omega$ be a unitary character of $F^\times$ with conductor $\pi^n\mathfrak{o}_F$. Let $\mathfrak{a}$ be an ideal of $\mathfrak{o}_F$ such that $\mathfrak{a}^2$ divides $\pi^n\mathfrak{o}_F$. Put $\mathfrak{b} = \mathfrak{a}^{-1}\pi^n\mathfrak{o}_F \subseteq \mathfrak{o}_F$.

(a) Show that there exists an element $c \in F$ such that $c\mathfrak{o}_F = \pi^n\mathcal{D}_F$ and also

$$\omega(1+t) = \psi(c^{-1}t)$$

for all $t \in \mathfrak{b}$. [*Hint*: Suppose that $\mathfrak{a} \neq \mathfrak{o}_F$, so that $\pi$ divides $\mathfrak{a}$. If $t, z \in \mathfrak{b}$, then $tz \in \pi^n\mathfrak{o}_F$. Consequently, $\omega(1+t)\omega(1+z) = \omega(1+t+z)$, and therefore the map that sends $t$ to $\omega(1+t)$ is a character of the additive group $\mathfrak{b}$, and this extends to one on $F$. Now appeal to the isomorphism of $F$ with its dual.]

(b) For $c$ as in the previous part and $F$ now assumed to have characteristic zero, show that

$$W(\omega) = N(\mathfrak{b}\mathfrak{a}^{-1})^{-1/2} \sum_{x \in (1+\mathfrak{a})/(1+\mathfrak{b})} \overline{\omega}(c^{-1}x)\psi(c^{-1}x).$$

NOTE. When $\mathfrak{a} = \mathfrak{o}_F$, this result is identical to [RV99, Eq. (7.6)].

(c) For $\omega$ unramified, show that $W(\omega) = \omega(\pi)^d$ if $\pi^d\mathfrak{o}_F = \mathcal{D}_F$.

(d) Let $E$ be a number field containing $W(\omega)$. Then show that for every place $v$ of $E$ not dividing the residual characteristic $p$ of $F$, we have $|W(\omega)|_v = 1$. Using the Artin product formula, conclude that if $E$ has a unique place $u$ above $p$, then also $|W(\omega)|_u = 1$.

(e) Suppose that $\pi^n\mathfrak{o}_F = \pi\mathfrak{a}$. Then, using part (b), show that $z = (\omega(c^{-1})W(\omega))^2$ lies in the cyclotomic extension $E = \mathbb{Q}(e^{2\pi i/p^r})$ for some $r \geq 1$. Show further, using part (d), that $z$ must be a root of unity.

(f) (Lamprecht, Dwork) Suppose that $\omega$ is either *unramified* (that is, $n = 0$) or *wildly ramified* (that is, $n \geq 2$). Show that $W(\omega)$ is a root of unity. [*Hint*: This is clear if $n$ is even. When $n$ is odd, write $\pi^n\mathfrak{o}_F = \pi\mathfrak{a}$ and apply the previous part.]

**Exercise 7.11.** Let $K$ be a global field. For any idele class character $\chi = (\chi_v)$, put

$$W(\chi) = \prod_v W(\chi_v).$$

(a) Show that $W(\chi)W(\chi^{-1}) = 1$. In particular, if $\chi$ is unitary, then $|W(\chi)| = 1$.

(b) Let $K$ be a number field, and let $\chi$ be quadratic or trivial. Conclude from part (a) that $W(\chi) = \pm 1$. [This number is called the *sign of the functional equation* of $L(s, \chi)$.] Show, moreover, that if $\chi$ is unramified everywhere, then

$$W(\chi) = \widetilde{\chi}(\mathcal{D}_K)$$

where $\widetilde{\chi}$ is the associated character of the class group $\mathrm{Cl}_K$ (cf. [RV99, Proposition 5-19]). [*Hint*: Use the previous problem.]

(c) Let $\chi$ be a quadratic idele class character, and let $E/K$ be the quadratic extension corresponding to the open subgroup $\ker(\chi)$ of $C_K$ by class field theory ([RV99, Theorem 6-6]). Note that $\zeta_E(s) = \zeta_K(s)L(s,\chi)$ and deduce the formula

$$\varepsilon(s,\chi) = \left(|d_K|/|d_E|^{1/2}\right)^{1-2s}$$

where $d_K$ and $d_E$ are the discriminants of $K$ and $E$, respectively.

(d) (Hecke's Theorem, Serre's Proof) Let $K$ be a number field. Then prove:

THEOREM (Hecke). *The ideal class of $\mathcal{D}_K$ is a square in the class group* $\mathrm{Cl}_K$.

[*Hint*: Observe that it suffices to show that $\widetilde{\chi}(\mathcal{D}_K) = 1$ for every quadratic character $\widetilde{\chi}$ of $\mathrm{Cl}_K$. Then appeal to parts (b) and (c) of this problem.]

**Exercise 7.12.** We consider here the Fourier transforms of Schwartz–Bruhat functions.

(a) Let $F$ be a local field. Show that for every $f \in \mathcal{S}(F)$, its Fourier transform likewise lies in $\mathcal{S}(F)$. [*Hint*: For $F$ Archimedean, this is a well-known classical fact. In the non-Archimedean case, use that $f$ is a linear combination of characteristic functions of the basic compact sets $P^n$, where $P$ is the unique prime associated with $F$.]

(b) Let $K$ be a global field and assume that $f \in \mathcal{S}(\mathbb{A}_K)$. Show that the Fourier transform of $f$ is likewise in $\mathcal{S}(\mathbb{A}_K)$. [*Hint*: First prove that $\mathcal{S}(\mathbb{A}_K)$ is generated by factorizable functions $\bigotimes_v f_v$, where each $f_v$ lies in $\mathcal{S}(K_v)$ for all $v$, and then use the previous part to establish the result in this special case.]

(c) Use the results of [RV99, Chapter 3] to show that the Fourier transform map $\mathcal{S}(\mathbb{A}_K) \to \mathcal{S}(\mathbb{A}_K)$ extends to an isometry $L^2(\mathbb{A}_K) \to L^2(\mathbb{A}_K)$.

**Exercise 7.13.** (Hecke Characters) Let $K$ be a number field of degree $d$, let $S$ be a finite set of places containing $S_\infty$, and let $J_K(S)$ be the group of fractional ideals prime to $S$. Furthermore, let $\mathbb{I}_K(S)$ denote the subgroup of $\mathbb{I}_K$ consisting of ideles $y = (y_v)$ such that $y_v = 1$ at every place $v$ in $S$.

(a) Define $\alpha\colon \mathbb{I}_K(S) \to J_K(S)$ by

$$\alpha(y) = \prod_{v \notin S} P_v^{v(y_v)}.$$

Show that $\alpha$ is a surjective homomorphism with kernel

$$U(S) = \{y \in \mathbb{I}_K(S) : y_v \in \mathfrak{o}_v^\times \ \forall v \text{ finite}\}.$$

Conclude that every character $\chi$ of $\mathbb{I}_K$ that is unramified outside $S$ defines a character $\widetilde{\chi}$ of $J_K(S)$.

(b) Let $\beta$ be a *Grössencharakter* (or *Hecke character*) of $K$, which is to say a homomorphism from $J_K(S)$ to $\mathbb{C}^\times$ for which there exists an integral ideal $M$ with support $S$, complex numbers $s_1, \ldots, s_d$ and integers $m_1, \ldots, m_d$ such that for every $\alpha \in K^\times(M)$, one has

$$\beta((\alpha)) = \prod_{j=1}^{d} \sigma_j(\alpha)^{m_j} |\sigma_j(\alpha)|^{s_j}$$

where $\{\sigma_j\}$ is the set of embeddings of $K$ in $\mathbb{C}$. Show that every $\beta$ is of the form $\widetilde{\chi}$ for some character $\chi\colon \mathbb{I}_K \to \mathbb{C}^\times$ that is trivial on $K^\times$. [*Hint*: Show that $\beta$ defines a character of the ray class group $\mathrm{Cl}_K(M)$ and then lift it to the idele class group $C_K$.]

(c) Let $\Phi = \widetilde{\chi}$ for an idele class character $\chi$ unramified outside $S$. Show that $\Phi$ is in fact a Grössencharakter. [*Hint*: Take

$$M = \prod_{v \in S} P_v^{\max(1, n_v)}$$

where $n_v$ is the exponent of the conductor of $\chi_v$. For $\alpha \in K^\times(M)$, show that $\Phi((\alpha)) = \prod_{v \notin S} \chi_v(\alpha) = 1/(\chi_\infty \circ \sigma)(\alpha)$ by using first that $|\alpha|_v = 1$ for all $v$ in $S$ and second that $\chi$ is trivial on $K^\times$.]

**Exercise 7.14.** Fix an integer $k \geq 1$, and consider the *polylogarithm* function

$$l_k(z) = \sum_{n \geq 1} \frac{z^n}{n^k}.$$

Note that $l_1(z) = -\log(1-z)$. The special case of $l_2(z)$ is called *Euler's dilogarithm* function.

(a) Show that the series is normally convergent in the open unit disk $\{|z| < 1\}$ and that it has finite limit on the unit circle $S^1$ (respectively, $S^1 \smallsetminus \{1\}$) if $k > 1$ (respectively, $k = 1$).

(b) Let $\chi$ be a Dirichlet character of conductor $m \geq 1$. For $w$ an $m$th root of unity, put

$$G(\chi, w) = \sum_{c \bmod m} \chi(c) w^c.$$

Let $L(s, \chi)$ be the Dirichlet $L$-series, and assume that $\chi$ is not the trivial character if $k = 1$. Put $\chi(-1) = (-1)^r$. Then show that

$$L(k, \chi) = \frac{1}{m} \sum_{1 \leq b \leq m/2} G(\chi, e^{-2\pi i b/m}) \big[ l_k(e^{2\pi i b/m}) + (-1)^r l_k(e^{-2\pi i b/m}) \big].$$

[*Hint*: Expand $\chi$ in terms of the basis $\{e^{2\pi i b/m} : 0 \leq b < m\}$ of $\mathbb{C}[\mathbb{Z}/m\mathbb{Z}]$.]

(c) Show that

$$G(\chi, e^{-2\pi i b/m}) = \overline{\chi}(b) G(\chi, e^{-2\pi i/m}).$$

Simplify part (b) accordingly for $k = 1$ and deduce [RV99, Proposition 7-26].

(d) For any $k \geq 1$, show that there is an elementary expression for the Dirichlet series $L(k, \chi)$ if $\chi(-1) = (-1)^k$.

**Exercise 7.15.** Let $\chi$ be a Dirichlet character of conductor $m \geq 1$. As above, put

$$W(\chi) = \varepsilon(\chi |\cdot|^{1/2}) \quad (= \varepsilon(\tfrac{1}{2}, \chi)).$$

Show that

$$W(\chi) = \frac{G(\chi)}{\sqrt{m}} (-i)^r$$

where $r = 0$ (respectively, 1) if $\chi(-1) = 1$ (respectively, $-1$), and

$$G(\chi) = \sum_{a \bmod m} \chi(a) e^{2\pi i a/m}.$$

**Exercise 7.16.** Let $K$ be a number field and let $S$ be a set of primes in $\mathfrak{o}_K$. One says that $S$ has *lower Dirichlet density*

$$\underline{\delta}(S) = \liminf_{s \to 1^+} \frac{1}{\log \frac{1}{s-1}} \cdot \sum_{P \in S} \frac{1}{N(P)^s}$$

provided that the indicated lim inf exists. The *upper Dirichlet density* $\overline{\delta}(S)$ is likewise defined via the lim sup.

(a) Show that every $S$ has both a lower and an upper Dirichlet density. [*Hint*: Use [RV99, Proposition 7-31], part (ii), and also that a bounded sequence in $\mathbb{R}$ has an infimum and a supremum.]

(b) Show that $S$ has a Dirichlet density (as defined previously) if and only if $\underline{\delta}(S) = \overline{\delta}(S)$.

(c) Let $T$ be any finite set of primes in $\mathfrak{o}_K$. Show that $\underline{\delta}(S \cup T) = \underline{\delta}(S)$ and $\overline{\delta}(S \cup T) = \overline{\delta}(S)$. Conclude, in particular, that any finite set of primes has density zero.

(d) Let $\mathscr{P}_K^1$ denote the set of degree-one primes of $K$; that is, prime ideals $P$ of $\mathfrak{o}_K$ such that $N(P)$ is a rational prime. Show that for any set $S$ of primes, $\underline{\delta}(S \cap \mathscr{P}_K^1) = \underline{\delta}(S)$ and $\overline{\delta}(S \cap \mathscr{P}_K^1) = \overline{\delta}(S)$. Conclude that the set of primes in $\mathfrak{o}_K$ of degree greater than one has density zero.

**Exercise 7.17.** Let $E/K$ be a finite abelian extension of number fields with Galois group $G = \mathrm{Gal}(E/K)$. Let $U = N_{E/K}(C_E)$. Recall that $C_K/U$ is isomorphic to $G$ via class field theory. (See [RV99, Theorem 6-6].) For any character $\chi$ of $G$, let $\widetilde{\chi}$ denote the corresponding character of $C_K$ that is trivial on $U$. Keeping in mind that the extension $E/K$ is abelian, for any prime $P$ of $K$, we let

$$\varphi_P = \left(\frac{E/K}{P}\right)$$

denote the corresponding Frobenius element. Now set

$$\chi(P) = \begin{cases} \chi(\varphi_P) & \text{if } P \text{ is unramified in } E \text{ or if } \chi \text{ is trivial,} \\ 0 & \text{otherwise,} \end{cases}$$

and put

$$L(s,\chi) = \prod_P (1 - \chi(P)N(P)^{-s})^{-1}.$$

(a) Show that $L(s,\chi)$ converges absolutely in $\{\mathrm{Re}(s) > 1\}$ and further admits a meromorphic continuation to the whole $s$-plane, with no poles except possibly a simple one at $s = 1$, which occurs only when $\chi = 1$. Show, moreover, that there is a functional equation relating $s$ to $1 - s$. [*Hint*: Use [RV99, Theorems 6-6 and 7-16].]

(b) Show that

$$\zeta_E(s) = \prod_{\chi \in \widehat{G}} L(s,\chi).$$

[*Hint*: Write the left-hand side as an Euler product over the primes $P$ of $K$.]

(c) Show that

$$\lim_{s \to 1^+} \frac{1}{\log \frac{1}{s-1}} \sum_{P \in \mathscr{P}_K^1} \frac{\chi(P)}{N(P)^s} = \begin{cases} 1 & \text{if } \chi = 1, \\ 0 & \text{otherwise.} \end{cases}$$

The following four problems lead the reader through a proof of a version of the Tchebotarev density theorem, here reformulated in terms of Dirichlet density. Recall that $\mathscr{P}_{K,f}$ denotes the set of finite places of $K$, or, equivalently, the set of prime ideals of $\mathfrak{o}_K$.

> THEOREM. *Let $E/K$ be a finite Galois extension of number fields with Galois group $G$, and let $C$ be a conjugacy class in $G$. Set*
>
> $$S_K(C) = \left\{P \in \mathscr{P}_{K,f} : \left(\frac{E/K}{P}\right) = C\right\}.$$
>
> *Then $\delta(S_K(C))$ exists and equals* $\mathrm{Card}(C)/\mathrm{Card}(G)$.

We shall also need the degree-one version of $S_K(C)$, which we define as

$$S_K^1(C) = S_K(C) \cap \mathscr{P}_K^1.$$

The first problem is well known and deals with the abelian case. The next three encapsulate the ideas from an elegant proof due to O. Schreier.

**Exercise 7.18.** Let $E/K$ be a finite abelian extension, so that $C$ reduces to a singleton set $\{a\}$ for some $a \in G$. We recall from [RV99, Chapter 3, Exercise 13], the discrete form of the Fourier inversion formula: if $f$ is a complex-valued function on $G$, then

$$f(g) = \frac{1}{\operatorname{Card}(G)} \sum_{\chi \in \widehat{G}} \widehat{f}(\chi)\chi(g)$$

where

$$\widehat{f}(\chi) = \sum_{g \in G} f(g)\overline{\chi}(g).$$

(a) Given $a \in G$, define $f_a \colon G \to \mathbb{C}$ by

$$f_a(g) = \sum_{\chi \in \widehat{G}} \chi(a^{-1}g).$$

Show that $f_a$ is in fact the characteristic function of $\{a\}$.

(b) For all $g \in G$ show that

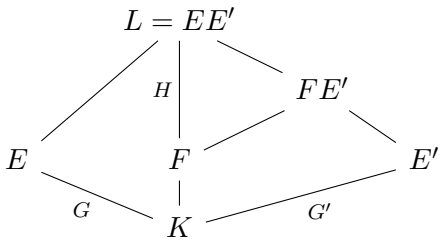
$$\sum_{P \in S_K(C)} \frac{1}{N(P)^s} = \sum_{g \in G} \sum_{P \in S_K(C)} \frac{f_a(g)}{N(P)^s} = \frac{1}{\operatorname{Card}(G)} \sum_{\chi \in \widehat{G}} \sum_{P} \frac{\chi(P)}{N(P)^s}.$$

(c) Show that $\delta(S_K(C))$ exists and equals $1/\operatorname{Card}(G) = \operatorname{Card}(C)/\operatorname{Card}(G)$. [*Hint*: Use the previous exercise.]

**Exercise 7.19.** Suppose that $E/K$ is any finite normal extension of number fields. Let $G$ be the associated Galois group, and let $C$ be a conjugacy class in $G$. Suppose further that $E'$ is an auxiliary finite abelian extension of $K$ with Galois group $G'$ such that $E$ and $E'$ are linearly disjoint over $K$. Put $L = EE'$.

(a) Show that $L/K$ is normal with Galois group $\mathcal{G} = G \times G'$.

(b) Choose any conjugacy class $C' = \{a\}$ in $G'$, and let $H$ be the subgroup generated by $CC'$ in $\mathcal{G}$. Put $F = L^H$, and consider the compositum $FE'$. We have the following diagram:




Show that $FE'$ is a normal extension of $F$ with Galois group given by

$$\operatorname{Gal}(FE'/F) \cong H/(H \cap \langle C \rangle) \cong \langle C' \rangle \subseteq G'.$$

Using the previous problem, conclude that

$$\delta\Big(\Big\{Q \in \mathscr{P}_F^1 : \Big(\frac{FE'/F}{Q}\Big) = C'\Big\}\Big) = \frac{1}{f'}$$

where $f' = \operatorname{Card}(\langle C' \rangle)$.

(c) For $P \in S_K^1(C)$, define $\alpha_P(F) = \mathrm{Card}(\{Q \in \mathscr{P}_F^1 : Q \mid P\})$. Show that

$$\alpha_P(F) = \frac{1}{\mathrm{Card}(H)} \mathrm{Card}(\{\tau \in \mathcal{G} : \tau C \tau^{-1} \in H\}).$$

[*Hint*: Use [RV99, Proposition 6-2].]

(d) Put $f = \mathrm{Card}(C)$. Assuming that $f \mid f'$, show for $P \in S_K^1(C)$ that

$$\left(\frac{L/K}{P}\right) \cap H = \varnothing.$$

**Exercise 7.20.** We continue in the context of the previous problem and always assume that $f \mid f'$.

(a) Show that

$$\lim_{s \to 1^+} \sum_{P \in S_K^1(C,C')} \frac{\alpha_P(F)}{N(P)^s} \sim \frac{1}{f'} \log\Big(\frac{1}{s-1}\Big)$$

where

$$S_K^1(C,C') = \Big\{P \in S_K^1(C) : \left(\frac{E'/K}{P}\right) = C'\Big\}.$$

(b) Show that

$$\delta(S_K^1(C,C')) = \frac{f}{nn'}$$

where $n$ and $n'$ denote $\mathrm{Card}(G)$ and $\mathrm{Card}(G')$, respectively. [*Hint*: Show that $\alpha_P(F) = nn'/(ff')$.]

(c) Let

$$n'_f = \mathrm{Card}(\{\sigma' \in G' : f \mid o(\sigma')\}).$$

Show that

$$\underline{\delta}(S_K(C)) \geq \frac{f n'_f}{nn'}$$

where again $\underline{\delta}$ denotes lower Dirichlet density, as defined above. [*Hint*: Any absolutely convergent summation indexed over $P \in S_K^1(C)$ can be rewritten as a double summation, first over $\sigma' \in G'$ and then over

$$\Big\{P \in S_K^1(C) : \left(\frac{E'/K}{P}\right) = \sigma'\Big\}.$$

Use this and appeal to part (b).]

**Exercise 7.21.** We continue in the context of the previous two problems.

(a) Show that we can choose $G'$ such that the quotient $n'_f/n'$ is arbitrarily close to 1. [*Hint*: Suppose that $f$ has prime factorization

$$f = \prod_{j=1}^{r} p_j^{a_j} \quad (a_j > 0).$$

Then take $G'$ of order

$$\prod_{j=1}^{r} p_j^{b_j}$$

such that for each $j$, $p_j^{b_j} - p_j^{a_j - 1}$ elements of $G'$ have order a power of $p_j^{a_j}$.]

(b) Use the previous exercise and the first part of this one to show that

$$\underline{\delta}(S_K(C)) \geq \frac{f}{n}.$$

(c) Show that

$$\underline{\delta}(S_K(G-C)) \geq \frac{n-f}{n}.$$

[*Hint*: $G - C$ is the disjoint union of conjugacy classes different from $C$.]

(d) Show that $\delta(S_K(C))$ exists and is given by $f/n = \mathrm{Card}(C)/\mathrm{Card}(G)$. [*Hint*: Use part (c) to conclude that

$$\overline{\delta}(S_K(C)) \leq \frac{f}{n}.$$

But the lower and upper Dirichlet densities always exist, and the latter bounds the former from above.]

**Exercise 7.22.** Let $K$ be a function field in one variable over a finite field $\mathbb{F}_q$. The first parts of this exercise lead to a proof of the following result:

> THEOREM. *The function $\zeta_K(s)$ is a rational function of $T = q^{-s}$. More precisely, there is a polynomial $P$ of degree $2g$, where $g$ is the genus of $K$, with $P(0) = 1$, such that*
>
> $$\zeta_K(s) = \frac{P(T)}{(1-T)(1-qT)}.$$
>
> *Furthermore, $P(1) = h$, the order of the divisor class group* $\mathrm{Pic}^0(K)$. (See [RV99, Section 7.5].)

REMARK. A celebrated theorem of A. Weil asserts in addition that $P(T)$ lies in $\mathbb{Z}[T]$ and admits a factorization over $\mathbb{C}$ of the form

$$\prod_{j=1}^{2g}(1-\alpha_j T)$$

with each $\alpha_j$ an algebraic integer of absolute value $q^{1/2}$. Consequently, every zero of $\zeta_K(s)$ lies on the line $\{\mathrm{Re}(s) = 1/2\}$—in other words, the *Riemann hypothesis* holds for $K$! But a proof of this result lies far beyond the scope of this book.

(a) Show that there exist infinitely many places $v$ of $K$ such that $N(v) = q$. Here the norm of $v$ simply measures the cardinality of the residue field associated with the corresponding local field $K_v$. [*Hint*: Suppose not. Then

$$\zeta_K(s) = \prod_{v \in S}(1-q^{-s})^{-1} \cdot \prod_{v \notin S}(1-q^{-f_v s})^{-1}$$

for a finite set $S$, with $f_v \geq 2$ for all $v$ outside $S$. Deduce from this that in fact, $\log \zeta_K(s)$ has a finite limit as $s \to 1^+$, and thus derive a contradiction to [RV99, Theorem 7-19].]

(b) Show that $\zeta_K(s)$ is a meromorphic function of $q^{-s}$. [*Hint*: Show that the image of $|\cdot|_{\mathbb{A}_K} : \mathbb{I}_K \to \mathbb{R}_+^\times$ is $q^{\mathbb{Z}}$, and then use that $\zeta_K(s) = L(|\cdot|_{\mathbb{A}_K}^s)$.]

(c) Using part (b) and [RV99, Theorem 7-19], show that $\zeta_K(s)$ takes the form given in the theorem above, with $P(q^{-s})$ an entire function.

(d) Using the functional equation for $\zeta_K(s)$, deduce the following functional equation for the numerator:

$$P(T) = q^g T^{2g} P\Big(\frac{1}{qT}\Big).$$

Deduce that $P$ must be a polynomial, and (by using [RV99, Theorem 7-22]) that $P(1) = h$.

(e) Let $\chi$ be any idele class character of $K$. Show that $L(\chi)$ is a rational function of $T = q^{-s}$. [*Hint*: Reduce to the unitary case and argue as above. Provided that $\chi \neq |\cdot|_{\mathbb{A}_K}^s$, the function $L(\chi)$ is in fact a polynomial in $T$.]

**Exercise 7.23.** Let $K$ be a number field. For $x > 0$, put

$$\pi_K(x) = \mathrm{Card}(\{P \text{ prime in } \mathfrak{o}_K : N(P) \leq x\}).$$

The "prime number theorem" for $K$ then asserts that as $x \to \infty$,

$$\pi_K(x) \sim \frac{x}{\log x}. \tag{7.21}$$

For $K = \mathbb{Q}$, this was conjectured by Riemann and proven independently by Hadamard and de la Vallée-Poussin in 1896. The object of this exercise is to deduce this theorem from the following theorem for Dirichlet series:

THEOREM (Tauberian Theorem for Dirichlet Series). *Let*

$$D(s) = \sum_{n \geq 1} \frac{a_n}{n^s}$$

*be a Dirichlet series with nonnegative coefficients $a_n$ that satisfies the following conditions:*

(i) *$D(s)$ is normally convergent in $\{\mathrm{Re}(s) > 1\}$.*

(ii) *$D(s)$ is invertible on the line $\{\mathrm{Re}(s) = 1\}$ except for a simple pole at the point $s = 1$ of residue $\alpha$.*

*Then*

$$\sum_{n \leq x} a_n \sim \alpha x$$

*asymptotically in $x$.*

(a) Show that

$$-\zeta_K'(s)/\zeta_K(s) = D(s) + \varphi(s)$$

for some function $\varphi(s)$ holomorphic and nonzero in $\{\mathrm{Re}(s) \geq 1\}$ and $D(s)$ the Dirichlet series defined by $a_n = \log n$, if $n$ is the norm of some prime $P$, and $a_n = 0$, otherwise.

(b) Using the results of [RV99, Section 7.7], show that the function $D(s)$ of the previous part satisfies the hypotheses of the Tauberian theorem and has residue 1 at $s = 1$. Deduce that

$$g(x) = \sum_{N(P) \leq x} \log N(P) \sim x.$$

(c) Put $b_n = a_n / \log n$. Show that for any $m \geq 1$, we have

$$\pi_K(m) = \sum_{n \leq m} \frac{a_n}{\log n} = \frac{g(m)}{\log m} + h(m)$$

where

$$h(m) = \sum_{n < m} g(n) \Big[ \frac{1}{\log n} - \frac{1}{\log(n+1)} \Big].$$

[*Hint*: $a_n = g(n) - g(n-1)$.]

(d) Using that $g(n) \sim n$, prove (7.21) by showing that $h(m)$ is $o(m/\log m)$; that is,

$$\lim_{m\to\infty} \frac{\log m}{m} h(m) = 0.$$

[*Hint*: Show first that

$$\Big(\frac{1}{\log n} - \frac{1}{\log(n+1)}\Big) \le \frac{1}{n \log^2 n}$$

for all $n > 1$.]

# Bibliography

[RV99] D. Ramakrishnan and R. J. Valenza, *Fourier Analysis on Number Fields*, Graduate Texts in Mathematics **186**, Springer-Verlag, New York, 1999.